CPE412 Pattern Recognition

Week 13

General Review

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PCA Algorithm

- Step-01: Get data.
- Step-02: Compute the mean vector (μ)(in some references $\bar{\mathbf{x}}$).
- Step-03: Subtract mean from the given data.
- Step-04: Calculate the covariance matrix.
- Step-05: Calculate the eigen vectors and eigen values of the covariance matrix.
- Step-06: Choosing components and forming a feature vector.
- Step-07: Deriving the new data set.

PCA-covariance matrix

μ

$$\bar{X} = \frac{\sum_{i=1}^{n} x_i}{n}$$

$$\text{Covariance Matrix} = \frac{\Sigma (x_i - \mu)(x_i - \mu)^t}{n}$$

covariance is a measure of the relationship between two random variables.

Problem-01:Solution

- Step-01: Get data.
- The given feature vectors are-
- x1 = (2.5,2.4)
- x2 = (0.5,0.7)
- x3 = (2.2, 2.9)
- x4 = (1.9, 2.2)
- x5 = (3.1, 3)
- x6 = (2.3, 2.7)
- x7 = (2,1.6)
- x8 = (1, 1.1)
- x9 = (1.5, 1.6)
- x10=(1.1,0.9)

x	y
2,5	2,4
0,5	0,7
2,2	2,9
1,9	2,2
3,1	3
2,3	2,7
2	1,6
1	1,1
1,5	1,6
1,1	0,9

Problem-01:Solution

- Step-02: Calculate the mean vector (μ).
- Mean vector (μ)
- = ((2.5 + 0.5 + 2.2 + 1.9 + 3.1 + 2.3+ 2 + 1 + 1.5 + 1.1) / 10, (2.4 + 0.7 + 2.9 + 2.2 + 3 + 2.7+ 1.6+ 1.1+ 1.6+ 0.9) / 10)
- = (1.81, 1.91)
- Thus,

$$\text{Mean vector } (\mu) = \begin{bmatrix} 1.81 \\ 1.91 \end{bmatrix}$$

Problem-01:Solution

- Step-03:Subtract mean vector (μ) from the given feature vectors.
  - x1 - μ = (2.5 -1.81,2.4-1.91) = (0.69,0.49)
  - x2 - μ = (0.5-1.81,0.7-1.91) = (-1.31, -1.21)
  - x3 - μ = (2.2-1.81, 2.9-1.91) = (0.39, 0.99)
  - x4 - μ = (1.9-1.81 , 2.2-1.91) = (0.09, 0.29)
  - x5 - μ = (3.1-1.81, 3-1.91) = (1.29, 1.09)
  - x6 - μ = (2.3-1.81, 2.7-1.91) = (0.49, 0.79)
  - x7 - μ = (2-1.81,1.6-1.91) = (0.19, -0.31)
  - x8 - μ = (1-1.81, 1.1-1.91) = (-0.81, -0.81)
  - x9 - μ = (1.5-1.81, 1.6-1.91) = (-0.31, -0.31)
  - x10 - μ =(1.1-1.81,0.9-1.91) = (-0.71, -1.01)

- Feature vectors (xi) after subtracting mean vector (μ) are-

$$\begin{bmatrix} 0.69 \\ 0.49 \end{bmatrix} \begin{bmatrix} -1.31 \\ -1.21 \end{bmatrix} \begin{bmatrix} 0.39 \\ 0.99 \end{bmatrix} \begin{bmatrix} 0.09 \\ 0.29 \end{bmatrix} \begin{bmatrix} 1.29 \\ 1.09 \end{bmatrix} \begin{bmatrix} 0.49 \\ 0.79 \end{bmatrix} \begin{bmatrix} 0.19 \\ -0.31 \end{bmatrix} \begin{bmatrix} -0.81 \\ -0.81 \end{bmatrix} \begin{bmatrix} -0.31 \\ -0.31 \end{bmatrix} \begin{bmatrix} -0.71 \\ -1.01 \end{bmatrix}$$

Problem-01:Solution

- Step-04:Calculate the covariance matrix.
- Covariance matrix is given by-

$$\text{Covariance Matrix} = \frac{\Sigma (x_i - \mu)(x_i - \mu)^t}{n}$$

Problem-01:Solution

$$m_1 = (x_1 - \mu)\,(x_1 - \mu)^t = \begin{bmatrix} 0.69 \\ 0.49 \end{bmatrix} \begin{bmatrix} 0.69 & 0.49 \end{bmatrix} = \begin{bmatrix} 0.48 & 0.34 \\ 0.34 & 0.24 \end{bmatrix}$$

$$m_2 = (x_2 - \mu)\,(x_2 - \mu)^t = \begin{bmatrix} -1.31 \\ -1.21 \end{bmatrix} \begin{bmatrix} -1.31 & -1.21 \end{bmatrix} = \begin{bmatrix} 1.72 & 1.59 \\ 1.59 & 1.46 \end{bmatrix}$$

$$m_3 = (x_3 - \mu)\,(x_3 - \mu)^t = \begin{bmatrix} 0.39 \\ 0.99 \end{bmatrix} \begin{bmatrix} 0.39 & 0.99 \end{bmatrix} = \begin{bmatrix} 0.15 & 0.39 \\ 0.39 & 0.98 \end{bmatrix}$$

$$m_4 = (x_4 - \mu)\,(x_4 - \mu)^t = \begin{bmatrix} 0.09 \\ 0.29 \end{bmatrix} \begin{bmatrix} 0.09 & 0.29 \end{bmatrix} = \begin{bmatrix} 0.01 & 0.03 \\ 0.03 & 0.08 \end{bmatrix}$$

$$m_5 = (x_5 - \mu)\,(x_5 - \mu)^t = \begin{bmatrix} 1.29 \\ 1.09 \end{bmatrix} \begin{bmatrix} 1.29 & 1.09 \end{bmatrix} = \begin{bmatrix} 1.66 & 1.41 \\ 1.41 & 1.19 \end{bmatrix}$$

$$m_6 = (x_6 - \mu)\,(x_6 - \mu)^t = \begin{bmatrix} 0.49 \\ 0.79 \end{bmatrix} \begin{bmatrix} 0.49 & 0.79 \end{bmatrix} = \begin{bmatrix} 0.24 & 0.39 \\ 0.39 & 0.62 \end{bmatrix}$$

$$m_7 = (x_7 - \mu)\,(x_7 - \mu)^t = \begin{bmatrix} 0.19 \\ -0.31 \end{bmatrix} \begin{bmatrix} 0.19 & -0.31 \end{bmatrix} = \begin{bmatrix} 0.04 & -0.06 \\ -0.06 & 0.10 \end{bmatrix}$$

$$m_8 = (x_8 - \mu)\,(x_8 - \mu)^t = \begin{bmatrix} -0.81 \\ -0.81 \end{bmatrix} \begin{bmatrix} -0.81 & -0.81 \end{bmatrix} = \begin{bmatrix} 0.66 & 0.66 \\ 0.66 & 0.66 \end{bmatrix}$$

$$m_9 = (x_9 - \mu)\,(x_9 - \mu)^t = \begin{bmatrix} -0.31 \\ -0.31 \end{bmatrix} \begin{bmatrix} -0.31 & -0.31 \end{bmatrix} = \begin{bmatrix} 0.10 & 0.10 \\ 0.10 & 0.10 \end{bmatrix}$$

$$m_{10} = (x_{10} - \mu)\,(x_{10} - \mu)^t = \begin{bmatrix} -0.71 \\ -1.01 \end{bmatrix} \begin{bmatrix} -0.71 & -1.01 \end{bmatrix} = \begin{bmatrix} 0.50 & 0.72 \\ 0.72 & 1.02 \end{bmatrix}$$

Problem-01: Solution

- Now, Covariance matrix
- = (m1 + m2 + m3 + m4 + m5 + m6+m7+m8+m9+m10) / 10
- On adding the above matrices and dividing by 10, we get-

$$\text{Covariance Matrix} = \frac{1}{10}\begin{bmatrix} 5.56 & 5.57 \\ 5.57 & 6.45 \end{bmatrix}$$

$$\text{Covariance Matrix} = \begin{bmatrix} 0.56 & 0.56 \\ 0.56 & 0.65 \end{bmatrix}$$

Problem-01: Solution

- Step-05: Calculate the eigen values and eigen vectors of the covariance matrix.
- λ is an eigen value for a matrix M if it is a solution of the characteristic equation $|M - \lambda| = 0$.
- So, we have-

$$\begin{vmatrix} 0.56 & 0.56 \\ 0.56 & 0.65 \end{vmatrix} - \begin{vmatrix} \lambda & 0 \\ 0 & \lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 0.56-\lambda & 0.56 \\ 0.56 & 0.65-\lambda \end{vmatrix} = 0$$

Problem-01: Solution

- From here,
- $(0.56 - \lambda)(0.65 - \lambda) - (0.56 \times 0.56) = 0$
- $0.36 - 0.56\lambda - 0.65\lambda + \lambda^2 - 0.31 = 0$
- $\lambda^2 - 1.21\lambda + 0.05 = 0$
- Solving this quadratic equation, we get $\lambda = 1.165, 0.045$
- Thus, two eigen values are $\lambda 1 = 1.17$ and $\lambda 2 = 0.05$.
- Clearly, the second eigen value is very small compared to the first eigen value. So, the second eigen vector can be left out.

Problem-01: Solution

- Eigen vector corresponding to the greatest eigen value is the principal component for the given data set.
- So. we find the eigen vector corresponding to eigen value $\lambda 1$.
- We use the following equation to find the eigen vector-
  - $MX = \lambda X$

Problem-01: Solution

- Eigen vector corresponding to the greatest eigen value is the principal component for the given data set.
- So. we find the eigen vector corresponding to eigen value $\lambda 1$.
- We use the following equation to find the eigen vector-
  - $MX = \lambda X$
  - where-
  - M = Covariance Matrix
  - X = Eigen vector
  - λ = Eigen value

Problem-01: Solution

- Substituting the values in the above equation, we get-

$$\begin{bmatrix} 0.56 & 0.56 \\ 0.56 & 0.65 \end{bmatrix} \begin{bmatrix} X1 \\ X2 \end{bmatrix} = 1.17 \begin{bmatrix} X1 \\ X2 \end{bmatrix}$$

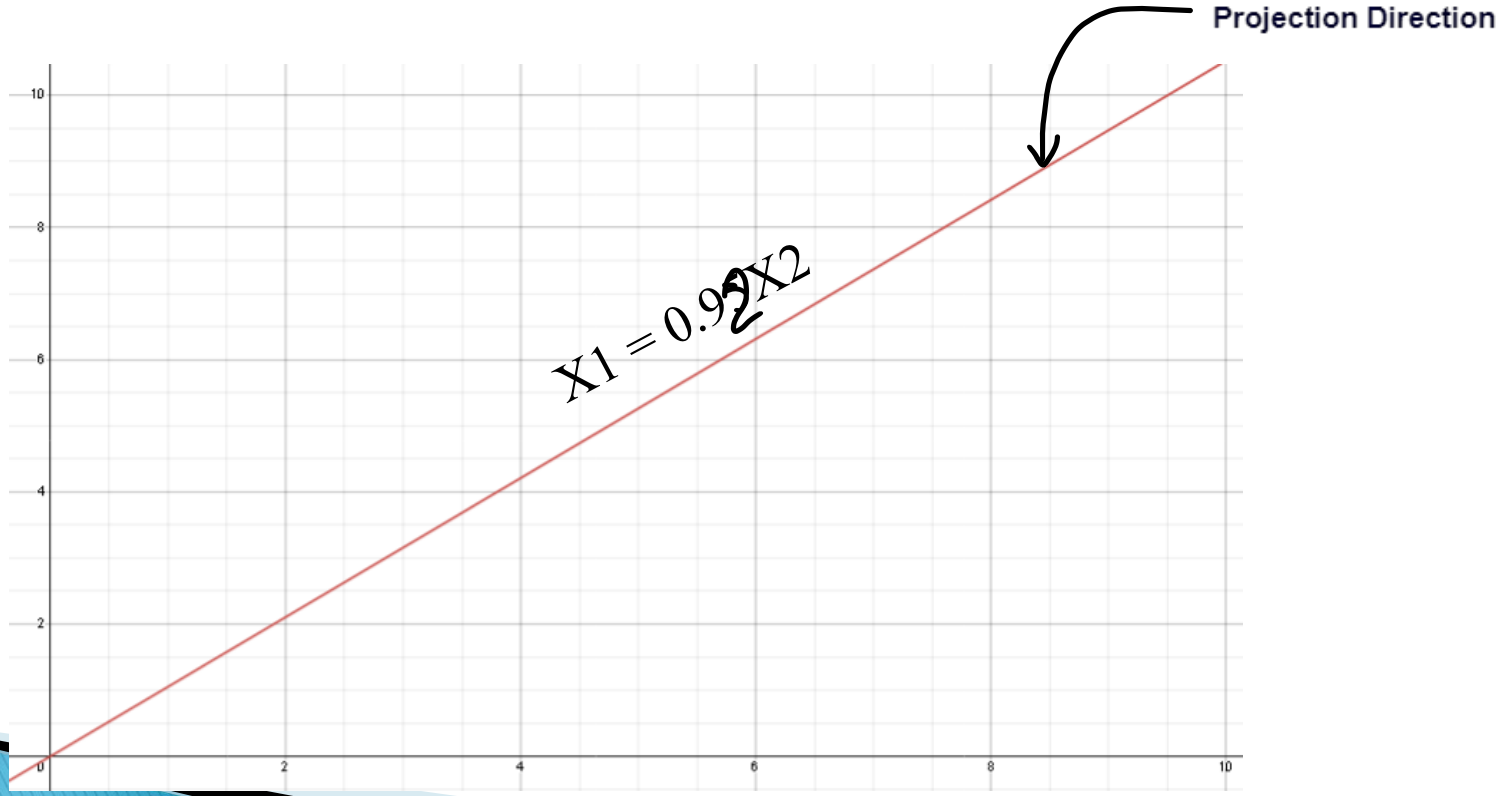
- Solving these, we get-
  - $0.56X1 + 0.56X2 = 1.17X1$
  - $0.56X1 + 0.65X2 = 1.17X2$
- On simplification, we get-
  - $0.61X1 = 0.56X2$ .....(1)
  - $0.56X1 = 0.52X2$ .....(2)
- From (1) and (2), $X1 = 0.92X2$
- From (2), the eigen vector is

take $X_1 = 0.52$
$X_2 = 0.56$

$$\text{Eigen Vector : } \begin{bmatrix} X1 \\ X2 \end{bmatrix} = \begin{bmatrix} 0.52 \\ 0.56 \end{bmatrix}$$

Problem-01: Solution

- Lastly, we project the data points onto the new subspace as-



After this slides just a help for solving equations

Perfect square equation

perfect squares

$$a^2 + 2ab + b^2 = (a+b)(a+b)$$

$$a^2 - 2ab + b^2 = (a-b)(a-b)$$

difference of squares

$$a^2 - b^2 = (a+b)(a-b)$$

Factoring perfect square trinomials

$$x^2 + 6x + 9$$

$$x^2 + 6x + 3^2$$

$$(x + 3)^2$$

$$(x + 3)(x + 3)$$

Solution of quadratic equation

Solution of quadratic equation($ax^2 + bx + c = 0$)is given by -

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \ldots (1)$$

Here, a = 1, b = 1 and c = 1 ($x^2 + x + 1 = 0$)

Just put the value of a, b and c in eq(1),

$$x = \frac{-1 \pm \sqrt{1^2 - 4*1*1}}{2*1}$$

$$x = \frac{-1 \pm \sqrt{-3}}{2}$$

$$\lambda^2 - 1.21\lambda + 0.05 = 0$$

$$\lambda = \frac{-(-1.21) \pm \sqrt{(-1.21)^2 - 4(1 * 0.05)}}{2 * 1}$$

$$\lambda = \frac{1.21 \pm \sqrt{1.46 - 0.2}}{2 * 1}$$

$$\lambda = \frac{1.21 \pm 1.12}{2}$$

$\lambda_1 = 1.165$

$\lambda_2 = 0.045$

Thanks ☺